Contralateral-Normalized Deformation and Stationary Log-Velocity Decomposition after Chronic Stroke: Geometric Characterization and Aphasia Prediction in the Aphasia Recovery Cohort

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Abstract

Nonlinear registration around a chronic stroke lesion reflects anatomical remodeling, premorbid asymmetry, and registration error rather than a uniquely identifiable physical displacement. We constructed a contralateral-normalized, lesional-only deformation proxy in the Aphasia Recovery Cohort and asked whether deformation or its stationary log-velocity Helmholtz–Hodge decomposition improves held-out prediction of Western Aphasia Battery Aphasia Quotient (WAB-AQ). After removing an affine transform fitted in the contralesional hemisphere, residual displacement was differenced from its mirrored homolog and summarized 3–20 mm outside the lesion. A smooth, positive-Jacobian embedding was mapped to an approximate stationary log-velocity; exponentiation error was audited before periodic-grid Helmholtz–Hodge decomposition. Ridge models were assessed with 20 repetitions of nested five-fold cross-validation. Of 214 deformation cases, 210 participants passed matching and quality control. All 214 log-velocity embeddings passed numerical quality control. The conventional lesion model achieved mean absolute error (MAE) 16.42 WAB-AQ points. Adding deformation yielded MAE 16.04 and a mean advantage of 0.38 points (95% participant-bootstrap interval -0.50 to 1.17). Adding Hodge features after lesion features yielded an advantage of -0.06 points (-0.42 to 0.29), and after deformation features -0.08 points (-0.35 to 0.21). Although a larger curl-free energy fraction was descriptively associated with lower AQ (Spearman $\rho = -0.240$, Holm $p = 0.002$), Hodge features did not improve prediction. Registration-derived displacement and log-velocity provide auditable geometric descriptors, but neither supports pressure estimation, and neither deformation feature family established robust incremental aphasia prediction.

Keywords: aphasia; chronic stroke; deformable registration; Helmholtz–Hodge decomposition; lesion-symptom mapping; stationary velocity

1 Introduction

Spatial normalization of brains with focal lesions is difficult because an intensity-driven transformation can deform intact anatomy to reduce mismatch at the lesion (Brett et al., 2001; Andersen et al., 2010). Lesion masking and filling can reduce this bias (Sdika and Pelletier, 2009), but a resulting displacement field still combines global pose, natural anatomical variation, chronic tissue loss, ventricular expansion, and algorithmic error. These qualifications are particularly important in chronic stroke, where lesion expansion and whole-brain shrinkage may continue years after the event (Seghier et al., 2014).

Deformation-based morphometry uses displacements and transformation Jacobians to characterize local shape and volume differences (Chung et al., 2001). In large-deformation diffeomorphic metric mapping (LDDMM), a diffeomorphism is generated by integrating a velocity field (Beg et al., 2005). A diffeomorphism near the identity can also be represented by a stationary vector-field logarithm and reconstructed through its exponential (Arsigny et al., 2006; Vercauteren et al., 2009). This log-domain representation permits vector-field analysis, but it is a registration parameter: it is not a measured tissue velocity and has no physical time unit.

The Helmholtz–Hodge decomposition separates a vector field into curl-free, divergence-free, and harmonic components. Boundary conditions determine the decomposition on a finite domain and must therefore be specified (Bhatia et al., 2013). Applied to a validated log-domain registration field, the decomposition may distinguish gradient-like and solenoidal geometric patterns. It does not, by itself, identify pressure. Recovering pressure or force from deformation requires constitutive material parameters, loads, and boundary conditions in a biomechanical inverse problem (Hogea et al., 2008).

The Aphasia Recovery Cohort (ARC) is an openly released chronic stroke imaging and behavioral resource (Gibson et al., 2024). Lesion anatomy strongly predicts aphasia severity, while tissue morphology beyond the lesion can contain additional information (Teghipco et al., 2024). We therefore asked three questions. First, can contralateral normalization yield a spatially explicit, lesional-only descriptor of chronic registration asymmetry? Second, do lesion-adjacent deformation summaries improve held-out WAB-AQ prediction beyond conventional lesion features? Third, do stationary log-velocity Helmholtz–Hodge descriptors explain additional variation or improve prediction after the displacement summaries? We used a participant-level bootstrap interval as the criterion for incremental predictive benefit and treated the log-velocity/Hodge extension as exploratory. All component–behavior correlations were descriptive secondary analyses.

2 Materials and Methods

2.1 Dataset, ethics, and cohort construction

ARC release 1.0.2 contains de-identified BIDS imaging and participant-level behavioral data from chronic stroke survivors (Gibson et al., 2024). Participants in the source studies provided informed consent under protocols approved by the University of South Carolina Institutional Review Board. The same board determined that the anonymized public release, with variables generalized to safe-harbor criteria, was exempt from further review (OpenNeuro Clinical, Pro00132576) (Gibson et al., 2024).

WAB-AQ is a 0–100 composite of spontaneous speech, auditory comprehension, repetition, and naming, with lower values indicating greater impairment (Kertesz, 2007). One T1-weighted acquisition was selected per participant and joined one-to-one to its WAB record by acquisition identifier. Duplicate cases, mixed method versions, and multiple retained acquisitions per participant were prohibited. The model cohort contained 82 participants reported as female and 128 reported as male. Sex was reported descriptively and was not added post hoc as a predictor.

2.2 Lesion delineation and anatomical processing

Lesion masks were generated with a full-resolution nnU-Net (Isensee et al., 2021) trained on ATLAS v2 (Liew et al., 2022). The training, inference, probability export, and lesion-quality-control source is publicly available in the pinned TR2StrokeSeg repository identified in the Data and Code Availability statement. Trained model weights and source images are external data dependencies and are not duplicated in this analysis repository.

Inpainted T1-weighted images provided a continuous intensity substrate for BrainSuite v23a processing (Shattuck and Leahy, 2002) and surface-constrained volumetric registration (SVReg) (Joshi et al., 2009). Generated lesion or inpainting-target intensities were never interpreted as biological tissue. The lesion, a 3-mm-dilated inpainting target, and their mirrored homologs were excluded from deformation summaries.

2.3 Contralateral-normalized deformation

Let $F(\mathbf{x})$ be the SVReg inverse coordinate map giving the subject coordinate corresponding to atlas voxel $\mathbf{x}$. On valid contralesional brain voxels more than two voxels from the midline, we fit

$$F_A(\mathbf{x}) = M\mathbf{x} + \mathbf{b} \tag{1}$$

by least squares with five iterations of median-absolute-deviation outlier rejection. The affine-removed residual in atlas-axis millimeters was

$$\mathbf{u}(\mathbf{x}) = M^{-1}\{F(\mathbf{x}) - F_A(\mathbf{x})\}. \tag{2}$$

Masked normalized Gaussian smoothing used a 2-mm kernel.

Let R reflect position through the symmetric atlas midline and reverse the left–right component of a vector. The lesion-associated asymmetry field was

$$\mathbf{u}_L(\mathbf{x}) = \mathbf{u}(\mathbf{x}) - R\mathbf{u}(R\mathbf{x}). \tag{3}$$

Both members of a mirrored pair had to lie inside valid atlas and mapped subject brain masks. Equation 3 was retained only in the lesional hemisphere; the reference hemisphere was stored as exact zeros.

Magnitude was $m = \|\mathbf{u}_L\|$. The gradient of the Euclidean distance outside the lesion defined outward unit normal $\mathbf{n}$, and radial displacement was $r = \mathbf{u}_L \cdot \mathbf{n}$ (positive outward and negative inward). Local volume change used $J = \det(I + \nabla\mathbf{u})$. For positive paired Jacobians, $\Delta \log J = \log J(\mathbf{x}) - \log J(R\mathbf{x})$. Summaries were computed in 3–5, 5–10, 10–20, and 20–40 mm shells. Prediction used eight fixed 3–20 mm summaries: median and 95th-percentile magnitude; median and mean-absolute radial displacement; outward and inward radial integrals; and positive and negative log-Jacobian integrals.

2.4 Stationary log-velocity and Helmholtz–Hodge decomposition

The lesional-only field in Equation 3 is a masked difference of registration displacements, not the original SVReg diffeomorphism. We therefore did not assume that it possessed a logarithm. Instead, we constructed and audited a smooth embedding before log-domain analysis. The field and valid mask were sampled on a 4-mm grid. Field values were multiplied by a raised cosine that increased from zero to one over 4 voxels (16 mm) inward from the valid-domain boundary, smoothed with a Gaussian kernel of 2.5 voxels (10 mm), and zero-padded by 6 voxels (24 mm) on each side. These fixed settings were applied to every case.

An embedding was eligible only if $\det(I + \nabla\mathbf{u}_e) > 0$ everywhere. We then approximated the stationary logarithm $\mathbf{v}$ satisfying

$$\exp(\mathbf{v}) \simeq I + \mathbf{u}_e \tag{4}$$

using six scaling-and-squaring steps and up to six symmetric residual corrections. Linear interpolation and identity continuation were used in field composition. Eligibility additionally required a positive Jacobian for the reconstruction and relative reconstruction RMSE no greater than 0.02. Thus, the analysis empirically tested whether a stationary representation adequately reconstructed each regularized input rather than treating the log as exact.

On the padded periodic grid, the Fourier transform $\widehat{\mathbf{v}}$ was decomposed for nonzero wave vector $\mathbf{k}$ as

$$\widehat{\mathbf{v}}_{\mathrm{cf}}(\mathbf{k}) = \frac{\mathbf{k}\,\mathbf{k}^{\mathsf{T}}}{\|\mathbf{k}\|^2}\widehat{\mathbf{v}}(\mathbf{k}), \tag{5}$$

$$\widehat{\mathbf{v}}_{\mathrm{df}}(\mathbf{k}) = \widehat{\mathbf{v}}(\mathbf{k}) - \widehat{\mathbf{v}}_{\mathrm{cf}}(\mathbf{k}), \tag{6}$$

where cf and df denote curl-free and divergence-free components. The zero frequency was assigned to the harmonic component. Energy fractions were the squared L^2 norm of each component divided by total stationary-velocity energy. Three predictors were retained: total stationary-velocity RMS, curl-free energy fraction, and divergence-free energy fraction. Decomposition reconstruction error and the sum of energy fractions were numerical audits.

The stationary field has displacement units over an arbitrary unit flow interval, not millimeters per second. Neither its divergence, its curl-free component, nor $\log J$ was labeled pressure. No elastic or poroelastic constitutive law, tissue stiffness, loading history, or mechanical boundary condition was available, so pressure was not an estimand.

2.5 Quality control and registration sensitivity

Eligible deformation cases required lesion laterality index $|V_L - V_R|/(V_L + V_R) \geq 0.80$, nonpositive-Jacobian fraction no greater than 0.05 for the original normalized field, and at least 1,000 valid voxels 3–20 mm from the lesion. When available, a direct non-inpainted SVReg field was normalized identically. The magnitude of the inpainted-minus-direct difference was recorded as registration sensitivity, not biological deformation. A sensitivity model added median registration sensitivity and contralateral affine-fit RMSE.

2.6 Prediction models

The intercept-only benchmark predicted the training-fold mean. The clinical model contained age at stroke and $\log(1 + \text{WAB days})$. The conventional lesion model added total lesion volume, left and right language-network lesion burden, and lesion laterality. The primary incremental model added the eight deformation summaries. Hodge models added the three log-domain descriptors to the conventional lesion model, with and without the displacement summaries. Sensitivity models added soft-lesion and boundary-uncertainty features, with and without deformation.

Within each training fold, missing predictors were median-imputed, standardized, and fit by ridge regression. A four-fold inner loop selected the penalty from 17 log-spaced values between 10^{-4} and 10^4 by MAE. A shuffled five-fold outer loop was repeated 20 times with identical splits for all models. Every preprocessing and selection operation was repeated inside the outer loop to limit selection bias (Varma and Simon, 2006).

2.7 Statistical analysis

Performance measures were out-of-fold MAE, RMSE, R^2, and Pearson r. A participant bootstrap with 5,000 resamples gave 95% intervals for the repeat-averaged absolute MAE of each model.

The primary incremental estimand was each participant's absolute error under the conventional model minus error under the deformation model, averaged over repeats; positive values favor the added feature family. A separate 5,000-resample participant bootstrap estimated its 95% interval. Predictive superiority required this interval to exclude zero. Paired Wilcoxon tests were also reported and Holm-adjusted across comparisons with the conventional model (Holm, 1979); they were not the superiority criterion because they test a rank-based rather than mean contrast.

Twelve descriptive Spearman correlations evaluated lesion volume, displacement, registration sensitivity, and Hodge descriptors. Participant bootstrap intervals used 5,000 resamples, and p values were Holm-adjusted across all 12 correlations. A sensitivity analysis repeated the primary and Hodge model comparisons after excluding the single right-dominant case. ARC aphasia subtype is derived from WAB performance and was incomplete and imbalanced in the analysis table; it was therefore not treated as an independent prediction target. No additional task or subtype model was fit post hoc.

3 Results

3.1 Cohort and displacement characteristics

The deformation manifest contained 214 unique cases, of which 211 matched the clinical table. After deformation quality control, 210 participants remained: 209 with left-dominant and 1 with right-dominant lesions. Median age at stroke was 58.5 years (IQR 50.0–66.0), median WAB assessment delay was 721 days (404–1658), median WAB-AQ was 69.1 (40.8–90.0), and median lesion volume was 79.1 mL (34.5–154.7).

Median 3–20 mm displacement magnitude was 6.20 mm (IQR 5.32–7.84), signed radial displacement was 0.16 mm (-0.67–0.99), and mean-absolute radial displacement was 3.53 mm (2.95–4.48). Median nonpositive-Jacobian fraction in the original normalized field was 1.49% (1.30–1.66%). Median direct-versus-inpainted registration sensitivity was 4.79 mm (3.64–6.23), and the proxy-to-sensitivity ratio was 1.34 (1.00– 1.71).

Figure 1 shows M2238, selected during method development as a high-asymmetry technical example rather than a representative participant.

3.2 Log-velocity and Hodge quality control

All 214 regularized embeddings, including all 210 cases in the prediction cohort, passed the positive-Jacobian and exponentiation criteria. Within the analysis cohort, the median minimum input Jacobian was 0.761 (IQR 0.695–0.806), and median relative exponentiation RMSE was 0.011 (0.010–0.015).

Median stationary-velocity RMS was 2.16 mm (IQR 1.78–2.65). Curl-free, divergence-free, and harmonic components accounted for median energy fractions of 0.284 (0.245–0.319), 0.692 (0.659–0.728), and 0.026 (0.017–0.033), respectively. These fractions characterize the stated tapered periodic embedding, not an anatomically unique decomposition.

3.3 Descriptive associations with WAB-AQ

Lesion volume had the largest descriptive association with WAB-AQ ($\rho = -0.698$, 95% bootstrap interval -0.769 to -0.612; Holm $p =< 0.001$). Greater displacement magnitude ($\rho = -0.352$, -0.470 to -0.222) and absolute radial displacement ($\rho = -0.368$, -0.492 to -0.239) were also associated with

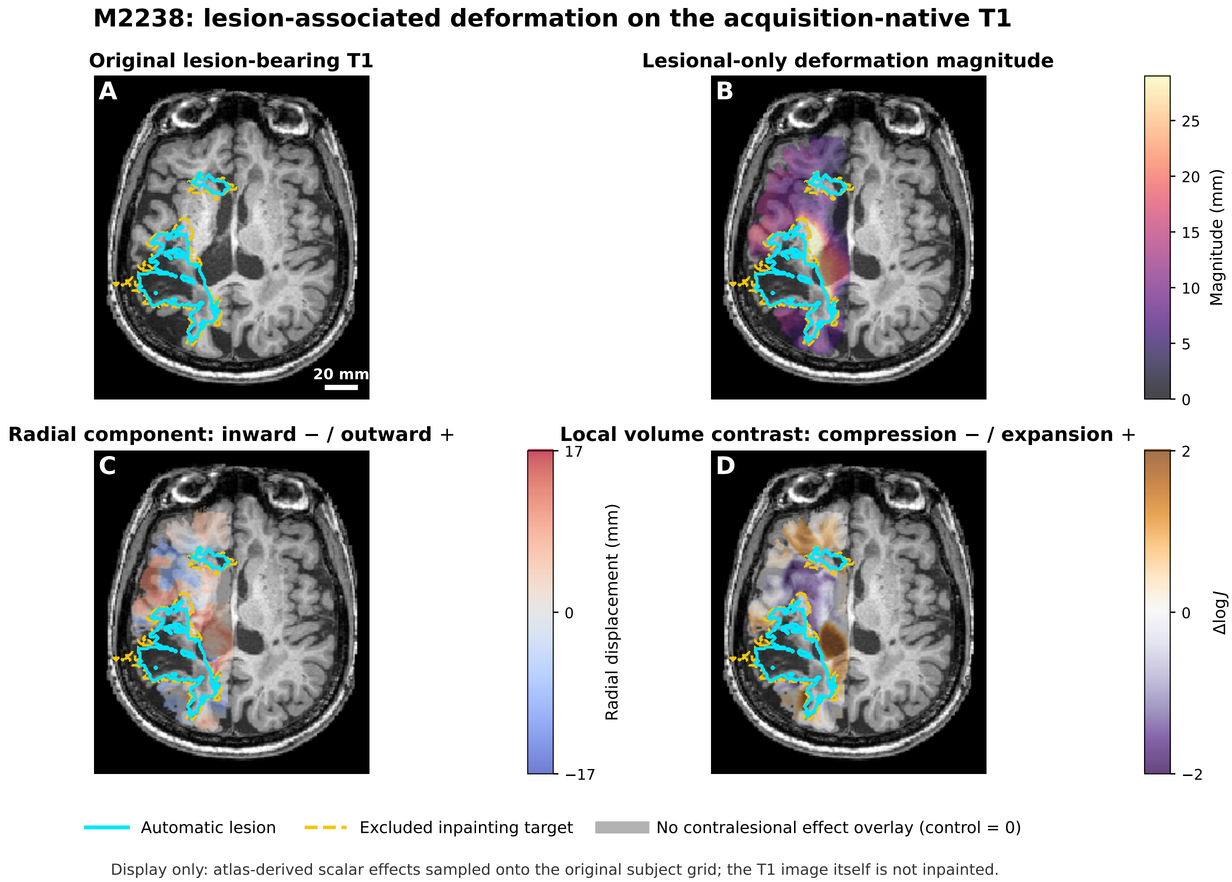

Figure 1: **Construction of the lesion-associated displacement proxy in M2238.** (A) Acquisition-native T1-weighted image with lesion (cyan) and excluded inpainting target (yellow dashed). (B) Lesional-only displacement magnitude. (C) Radial displacement (negative inward, positive outward). (D) Lesional-minus-mirrored contralesional log-Jacobian. The preselected case illustrates field construction and is not evidence of pressure.

lower AQ (both Holm $p < 0.001$). Signed radial displacement had a smaller negative association ($\rho = -0.235$, -0.356 to -0.112; Holm $p = 0.002$).

Total stationary-velocity RMS was not associated with AQ ($\rho = 0.071$, -0.063 to 0.206; Holm $p = 0.613$). A larger curl-free energy fraction was associated with lower AQ ($\rho = -0.240$, -0.365 to -0.104; Holm $p = 0.002$), whereas a larger divergence-free fraction was positively associated with AQ ($\rho = 0.176$, 0.038 to 0.303; Holm $p = 0.032$). These correlations were descriptive and were not adjusted for lesion features. Figure 2 displays the AQ associations; all 12 specified correlations are reported in the supplement.

3.4 Held-out WAB-AQ prediction

The intercept-only benchmark had MAE 24.14 (95% participant-bootstrap interval 22.28–25.97), and the clinical model had MAE 24.23 (22.36–25.99). The conventional lesion model improved MAE to 16.42 (14.87– 18.09), with RMSE 20.36, $R^2 = 0.454$, and Pearson $r = 0.675$ (Table 1; Fig. 3).

Adding displacement summaries yielded MAE 16.04 (14.46–17.72). The primary mean MAE advantage was 0.38 points (95% interval -0.50 to 1.17), and 60.5% of participants had lower

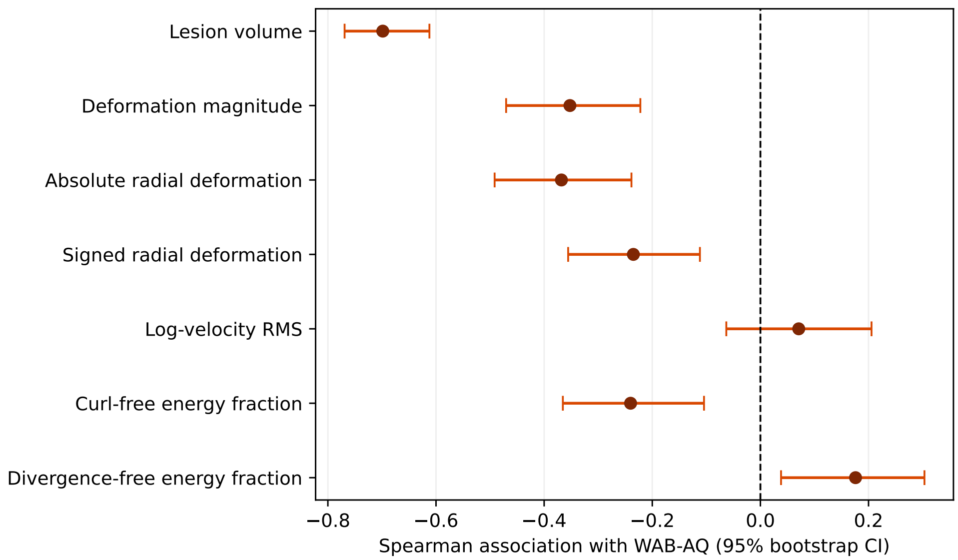

Figure 2: **Descriptive associations with WAB-AQ.** Points are Spearman correlations and bars are participant-bootstrap 95% intervals. Multiplicity was controlled by Holm adjustment across the complete family of 12 descriptive correlations. Component fractions depend on the tapered periodic boundary construction and are not estimates of pressure or material properties.

repeat-averaged absolute error. Although the Holm-adjusted signed-rank test was $p = 0.046$, the mean-advantage interval crossed zero; incremental predictive benefit was therefore not established.

Adding Hodge descriptors to the conventional model yielded MAE 16.48 and a mean advantage of -0.06 points (-0.42 to 0.29; Holm $p = 0.688$). Adding Hodge descriptors after the displacement summaries yielded MAE 16.12 and a direct advantage of -0.08 points (-0.35 to 0.21; $p = 0.573$). Thus, the descriptive component–AQ associations did not translate into incremental held-out prediction.

The lesion-uncertainty sensitivity model had MAE 15.71. Adding displacement features produced MAE 15.77 and a mean advantage of -0.06 points (-0.77 to 0.61; $p = 0.723$). In the left-dominant-only cohort ($n = 209$), the primary deformation advantage was 0.33 points (-0.51 to 1.09), consistent with the full-cohort conclusion.

4 Discussion

We constructed an auditable lesion-associated displacement proxy and extended it to a validated stationary log-velocity representation. The log-domain step implements the idea underlying diffeomorphic flow models: a velocity field is exponentiated to produce a deformation. It does not recover the time-dependent velocity optimized by LDDMM, because the input here is an affine-removed, mirrored-difference, one-sided field computed after registration. Requiring a positive-Jacobian smooth embedding and low exponentiation error makes the restricted stationary representation testable on each case.

The Hodge decomposition showed that the divergence-free component carried most energy under the chosen embedding, while a greater curl-free fraction was descriptively associated with worse aphasia. Two cautions constrain this observation. First, the curl-free and divergence-free components depend on the raised-cosine taper, smoothing scale, padding, and periodic boundary condition; they are not unique anatomical properties. Second, Hodge descriptors neither improved the conventional

Table 1: Out-of-fold performance averaged across 20 repeated outer-CV splits. MAE intervals use a participant bootstrap of repeat-averaged absolute errors. The same 210 participants and outer splits were used for every model.

Model	MAE (95% CI)	RMSE	R^2	Pearson r
Intercept only	24.14 (22.28, 25.97)	27.68	-0.009	-0.119
Clinical	24.23 (22.36, 25.99)	27.83	-0.020	-0.132
Conventional lesion	16.42 (14.87, 18.09)	20.36	0.454	0.675
Lesion + deformation	16.04 (14.46, 17.72)	20.33	0.456	0.677
Lesion + log-velocity Hodge	16.48 (14.88, 18.15)	20.50	0.447	0.670
Deformation + log-velocity Hodge	16.12 (14.57, 17.79)	20.35	0.454	0.676

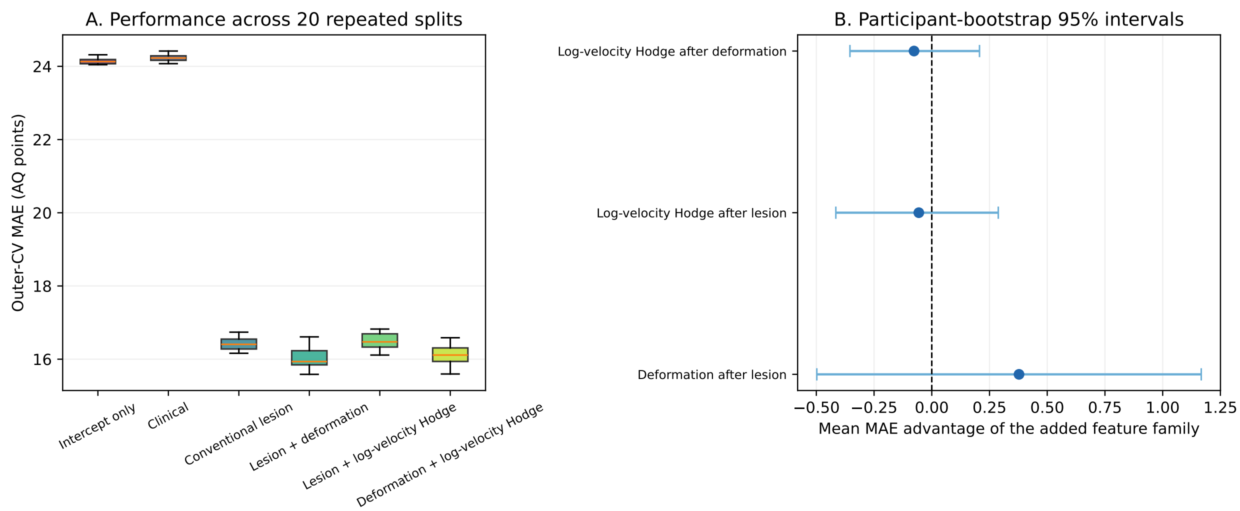

Figure 3: **Repeated nested-CV model comparison.** (A) MAE across 20 complete outer-CV repetitions. (B) Participant-level mean MAE advantages for the added feature family. Positive values favor the augmented model; bars are participant-bootstrap 95% intervals. Each incremental interval crosses zero.

lesion model nor added to the displacement model. The association may reflect lesion-linked geometry or residual registration structure rather than independent behavioral information.

The primary displacement analysis likewise yielded a small apparent MAE reduction, but its participant-bootstrap interval included zero. The signed-rank result does not alter that conclusion: it concerns ranks of paired errors, whereas the target estimand was their mean. Deformation also added no value after soft lesion and boundary-uncertainty features. Informational overlap is plausible because deformation integrals, lesion uncertainty, and lesion burden all reflect lesion extent and boundary complexity. Measurement noise is another plausible contributor: proxy magnitude was only modestly larger than direct-versus-inpainted registration sensitivity.

No result estimates intracranial or tissue pressure. Pressure has units of force per area, whereas the stationary field here has displacement units over an arbitrary unit interval. A pressure inversion would require, at minimum, a mechanical constitutive model, tissue-specific material parameters, reference configuration, loads, and boundary conditions. The single chronic T1-weighted scan supplies none of these. Divergence, curl-free energy, and log-Jacobian can be useful geometric features, but relabeling them as pressure would be a category error.

Several limitations remain. A single chronic image cannot identify premorbid anatomy or distinguish tissue loss, ventricular expansion, and chronic remodeling from acute displacement. The contralateral hemisphere is an internal reference, not a normative atlas, and the method is restricted to strongly unilateral lesions. The 10-mm smoothing used to obtain a common positive-Jacobian embedding limits Hodge analysis to coarse spatial structure. Upstream lesion delineation and inpainting can influence correspondences despite target exclusion. ARC focuses on aphasia and is almost entirely left-lesioned; no independent external cohort was available. WAB-AQ is a composite severity measure, and the current results do not generalize to individual language tasks, aphasia subtype, recovery, or treatment response. Repeated nested cross-validation reduces split dependence but does not replace external validation.

Future work could examine longitudinal displacement from acute to chronic imaging, compare alternative Hodge boundary conditions, and preregister external validation. A genuine pressure analysis would be a separate study requiring biomechanical imaging or experimentally justified material models and boundary conditions, not a renaming of registration features.

5 Conclusion

Contralateral normalization and a numerically validated stationary logarithm provide transparent geometric descriptions of chronic stroke registration asymmetry. In ARC, displacement and Hodge descriptors were associated with aphasia severity in unadjusted descriptive analyses, but neither established robust incremental WAB-AQ prediction beyond conventional lesion features. The analysis supports geometric characterization, not pressure estimation, physical tissue velocity, or causal mass effect.

Data and Code Availability

ARC is available from OpenNeuro as ds004884, release 1.0.2. The analysis code, tests, aggregate results, and manuscript sources are available in the ARCDeformationAnalysis repository. Lesion delineation training and inference code is available at the pinned TR2StrokeSeg revision. The analysis repository is self-contained from the documented derivative-input boundary; raw ARC data, trained nnU-Net weights, BrainSuite, participant-level derived images, joined clinical tables, and predictions are not redistributed.

Author Contributions

Anand A. Joshi: Conceptualization, Methodology, Software, Formal analysis, Visualization, Writing—original draft, and Writing—review and editing. This statement and the final author list must be confirmed before submission.

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